

MicroTraitLLM: A RAG-based, Microbe-focused LLM for Researchers

Singh, Kunvarkaranveer <ks4268@drexel.edu>; Allipilli, Shanmukha Rao <sa3935@drexel.edu>; Dawud-Sulaiman, Abdullah <aod27@drexel.edu>; Patel, Chirayu <cnp68@drexel.edu>; Nhu, Chris <cn527@drexel.edu>; Rogers, Glen <gr467@drexel.edu>

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Abstract (½ Page):

Rapid advances in Large Language Models (LLMs) and broad potential applications in biological research make them a compelling point of investigation. Key challenges remain, including the tendency of LLMs to hallucinate if they are not provided with domain-specific information, thereby potentially misinforming users; the small number of LLM applications directed to prokaryotes despite significant advances in microbiome studies, especially when compared to the many human-specific and eukaryotic domain-specific LLM application that have recently been released; and the inability of current microbe-specific LLM tools to provide sufficiently comprehensive, accurate, and timely answers with proper citations.

We present an improved version of MicroTraitLLM (MTLLM), a retrieval-augmented generation (RAG) LLM which utilizes zero-shot and single-shot prompting to give specific, citation-based answers for researchers through its connection to the live-updating PubMed Central Open Access article database. However, MicroTraitLLM suffers from two main issues: lack of validations and the underwhelming number of features compared to modern LLM agents. We therefore determined a focus on two main tasks: validation studies and general system improvements. We have integrated the following updates to the framework of MicroTraitLLM: 1.) LLM fine-tuning based on a curated microbial-text relevant dataset; 2.) Vision Question Answering (VQA) based on image-aware RAG; 3.) Article and citation validation studies; 4.) Validation studies in biomedical domain-specific LLMs; and 5.) Local Database storage, parsing, and analysis studies in preparation for future vector-summary comparison studies.

Materials and Methods:

Validation Studies

Model Comparison Study (Chirayu)

Our first validation study consisted of a model comparison study. Our main objective with this study was to determine if smaller, biomedical fine-tuned models could serve as a reasonable model within the MicroTraitLLM framework. The model comparison study compared four biomedical language models of different sizes and training backgrounds—BioGPT (347M parameters), BioMedLM (2.7B parameters), BioMedGPT-7B (7B parameters), and a GPT-2 baseline (124M–774M)—to evaluate their effectiveness in information extraction with regards to the microbial domain. The evaluation focused on three main parameters: Extraction accuracy, performance metrics, and efficiency metrics. Extraction accuracy predominantly focused on determining the number of organisms correctly identified, the number of traits recovered per organism, and the overall success rate defined as the proportion of queries yielding at least one correct organism. Performance measurements captured average and median latency per query, latency variability, total tokens generated, and average tokens per query, enabling a detailed view of responsiveness and output behavior. Finally, efficiency metrics were incorporated to assess computational performance relative to extraction quality. These included organisms extracted per second, memory utilization with support for 8-bit quantization, and explicit analysis of the trade-off between processing speed and extraction accuracy across differently sized models.

To ensure comprehensive evaluation, a 15-query benchmark dataset was developed covering a wide range of microbiological topics. It included seven major pathogens (e.g., tuberculosis, pneumonia, and foodborne pathogens), three specific foodborne organisms (*Salmonella*, *Listeria*, *Vibrio*), one environmental organism (*Bacillus subtilis*), one probiotic (*Lactobacillus*), one industrial genus (*Streptomyces*), and two model organisms (*E. coli* and *S. aureus*). Each query required extracting organism identity and a full spectrum of biological traits: morphological (Gram staining, cell shape), metabolic (oxygen use, fermentation), biochemical (enzymes and products), pathogenicity (virulence factors, toxins), and environmental attributes (habitat and growth conditions).

Article Validation and Citation Accuracy (AVCA) Study (Kunvarkaranveer and Shanmukha)

Our second validation study predominantly focused on article validation and citation accuracy (AVCA). Our primary objective with this study was to demonstrate that the answers provided by MicroTraitLLM are both topic-relevant to the user query (avoiding hallucinatory topics) as well as ensuring that the citations are accurate for transparency and clarity to the user.

The Article Validation and Citation Accuracy (AVCA) subsystem serves as the filtering and evaluation layer within the larger retrieval-to-generation pipeline. After the retrieval subsystem assembles a set of candidate documents, AVCA determines which articles are most relevant to the user's query and assigns graded confidence levels that can be used by downstream components. For evaluation purposes,

we selected the representative biological query “*antibiotic resistance in gram-negative bacteria*”, which reflects the domain of the larger research project.

To embed both the query and each document into a comparable numerical representation, AVCA uses the TF-IDF (Term Frequency–Inverse Document Frequency) model. TF-IDF increases the weight of terms that are frequent within an individual document while decreasing the weight of terms that occur commonly across the entire corpus. This allows the vectorization process to highlight biologically meaningful terms and reduce noise from ubiquitous scientific phrasing. The same TF-IDF vectorizer is applied to both the query and all candidate articles to maintain a consistent embedding space.

Formula:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \log(N / \text{DF}(t)) \text{ [3]}$$

Where:

- $\text{TF}(t, d)$ = term frequency of term t in document d
- N = total number of documents in the corpus
- $\text{DF}(t)$ = number of documents containing term t

Once TF-IDF vectors are generated, AVCA computes three similarity metrics for each document–query pair. Cosine similarity measures the angle between vectors and captures directional alignment of term usage patterns using $\cos(A, B) = (A \cdot B) / (\|A\| \|B\|)$. Dot product similarity evaluates the magnitude-weighted overlap between the TF-IDF vectors, emphasizing documents that strongly express key domain-specific terms present in the query using $\text{Dot}(A, B) = \sum_i (A_i \times B_i)$. Unlike cosine similarity, the dot product incorporates absolute TF-IDF weights, making it sensitive to term emphasis and document length. This helps differentiate documents that strongly emphasize key biomedical terms. Pearson correlation coefficient measures proportional linear agreement between term distributions, independent of vector magnitude, making it sensitive to similarities in overall pattern rather than absolute counts using $\text{Pearson}(A, B) = \text{Cov}(A, B) / (\sigma_A \sigma_B)$. Pearson can capture cases where two documents emphasize similar patterns of term usage even if absolute counts differ. Using all three metrics provides a more complete assessment of relevance and allows us to evaluate consistency across different similarity formulations.

After similarity scores are calculated, AVCA assigns each document to a confidence band (High, Medium, or Low) based on quantile thresholds. This percentile-based segmentation ensures that the banding strategy adapts to the statistical properties of the dataset rather than relying on arbitrary fixed cutoffs. The resulting framework provides a robust and interpretable method for ranking scientific articles and preparing them for the citation validation and LLM reasoning stages of the system.

Vector Ablation Study

One of the main weaknesses of the currently published material on MicroTraitLLM is that the information compilation methodology, which is mainly article summarization through LLMs, has not been shown to be more efficient than traditional vectorization methods. Therefore, to compare vectorization and summarization techniques, we conducted an ablation study to determine the best number of vectors (Top K) for information retrieval and relay. The answer creation system used the same

framework and system prompts as MicroTraitLLM’s base system, but the information architecture was changed to a vector-based system rather than a generated summary system. The PMCOAS remained as the main database for article parsing. The models that were used for the study were Llama3.2 (3 billion parameters) and Gemma2 (2 billion parameters).

The question subset used for evaluation during this set of testing consisted of 10 “basic” level questions, which have more concrete biological answers, and 19 “complex” level questions, which are more nuanced questions with regards to microbes. These questions were derivatives of questions previously outlined and tested with microbial experts, lending credence to their effectiveness in the microbiome field. Each question was evaluated with Top-K vector sizes ranging from zero to $K = 10$ vectors, and each of these vector sizes was given 100 iterations to ensure a comprehensive answer range. Therefore, approximately 19,000 answers were generated in the creation of this ablation study for evaluation for each model. Each answer was evaluated through a hybrid scoring system, which utilized a combined score of semantic similarity (as determined by cosine similarity) and ROUGE score. Since ROUGE measures a recall of n-grams, there is a chance that rephrased or alternative language that is correct for the microbial domain may be missed or weighted lower by the ROUGE score evaluation system. Therefore, we also incorporate the cosine similarity in an attempt to gauge a more appropriate response for answer “correctness” and mitigate the bias from the ROUGE score. However, not all outputs provided legitimate answers, with the occasional hallucination or missing response appearing within the dataset. Therefore, all answers with a hybridized score of 0.05 (5% correctness) were removed from the datasets.

System Improvements

Vision Question Answering (VQA) Prototype (Abdullah)

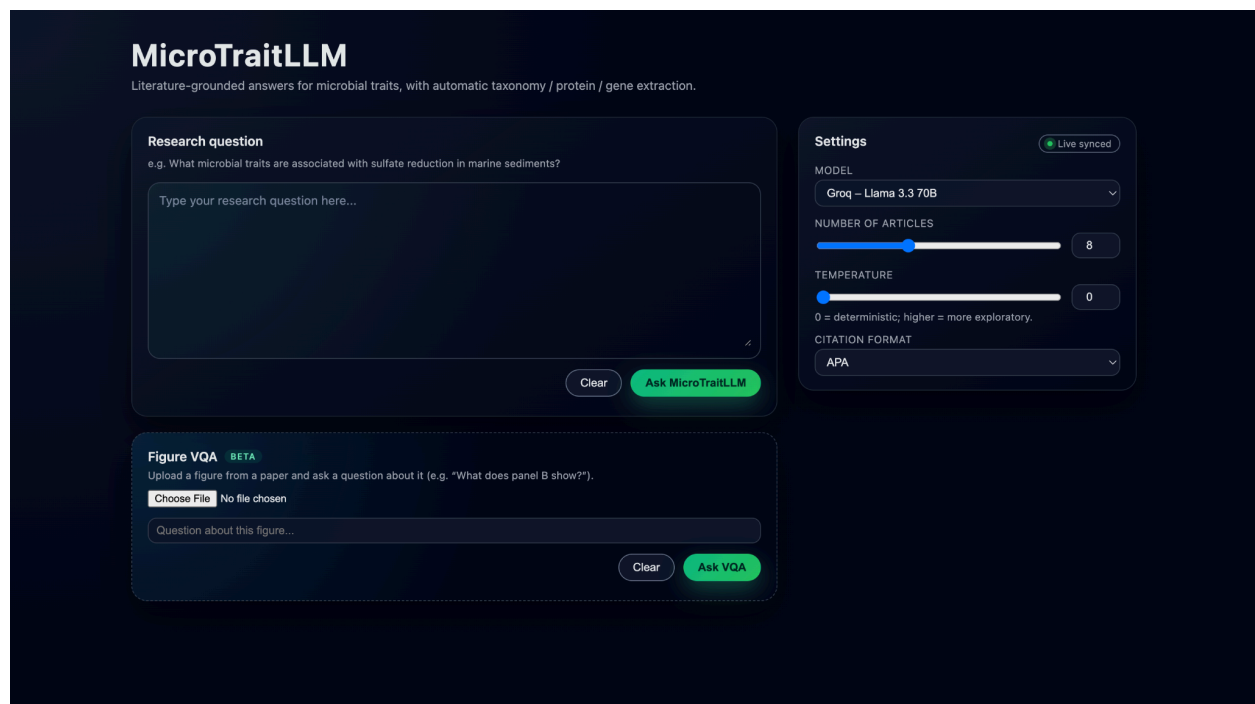
We evaluated the VQA integration qualitatively by testing it on a set of example figures and questions representative of typical microbiology literature. The primary goal of this first iteration was not large-scale benchmarking, but rather to validate that the end-to-end pipeline—from file upload, through the Flask backend, to the multimodal model and back to the UI—behaved reliably and produced scientifically sensible explanations.

One of the test cases used in development was a histogram of pixel intensities for an image file (e.g., unalltm1.tif). When the histogram figure was uploaded and the user asked “What does this figure show?”, the VQA system produced a structured description of the plot. The model correctly identified the x-axis as pixel intensity values (ranging roughly from 0–255) and the y-axis as a count or frequency of pixels. It highlighted the presence of a prominent peak around an intensity of ~100, noted the approximate count at the peak (on the order of a few thousand pixels), and described the tapering distribution as intensities increased toward the high end of the range. Importantly, the explanation emphasized that the image contained many more dark-to-midtone pixels than bright pixels, which is exactly the interpretation a human expert would derive from such a histogram. This confirms that the VQA pipeline is capable of translating low-level visual information into high-level, text-based interpretations that are directly usable by a researcher.

From a systems perspective, we also assessed robustness to API limitations. During development, we encountered rate limit and credit errors from the OpenAI vision endpoint. After implementing the fallback logic in `run_vqa`, the system gracefully degraded when OpenAI limits were exceeded: instead of crashing or

returning an opaque error, the code attempted to re-answer the question using a Groq vision model. When both providers failed (e.g., during early experiments with a decommissioned model ID), the system returned a clear error message explaining that VQA had failed across providers. This behavior is important for a research-facing tool, since it eliminates silent failures and reduces user confusion.

Although we did not yet perform a full quantitative evaluation of VQA accuracy across a standardized biomedical figure dataset, these initial experiments demonstrate that the current prototype can (1) correctly parse and interpret basic figures such as histograms, and (2) integrate cleanly into the existing MicroTraitLLM web interface. In the future, we plan to extend this work to more complex figure types (e.g., growth curves, bar charts with multiple conditions, and multi-panel figures) and to systematically evaluate the vision models against curated ground-truth descriptions.



Screenshot of the MicroTraitLLM web UI showing the main research question input and the new “Figure VQA” card with image upload and question fields.

Low Rank Adaption (LoRA) Fine Tuning (Chris)

We created a custom dataset set up as Question and Answer pairs with context to try and mimic as similarly as possible the end goal of the fine tuned model (which reads in articles as context, answers a question with an academic tone). This dataset consists of roughly 30 questions from different domains of microbiology, with real citations provided as context. After creating this dataset, an open source 7B parameter model called “Mistral” was finetuned using LoRA to produce a model that produces output in a specific format more akin to an Action Expert.

Database Optimization and Article Reranker (Glen)

To improve dataset reliability and performance, we elected to replace the previous data-access method used in MicroTraitLLM. Earlier iterations of the system relied on the NCBI BioC API to retrieve full-length articles from the PubMed Central Open Access Subset (PMCOAS)[1]. However, due to recent governmental changes and associated server outages, consistent access to these articles has become increasingly difficult. Consequently, the most practical solution was to obtain and host a complete local copy of the PMCOAS.

This transition required three primary steps. First, the entire dataset needed to be transferred via the File Transfer Protocol (FTP) system provided by NCBI. This transfer was accomplished using the AWS S3 protocol, and the full database (approximately 500 GB) is now stored locally for fast, reliable retrieval. All articles and their accompanying metadata are provided in XML format; however, MicroTraitLLM was originally designed to ingest JSON inputs. To address this discrepancy, the new data-parsing subsystem was optimized for XML, and the majority of MicroTraitLLM's internal operations now natively utilize XML rather than JSON. The system also includes an automated update mechanism to refresh the dataset, ensuring continued access to current PMCOAS content. Overall, the shift to a locally hosted database provides substantially faster, more stable data access and improves the reproducibility of system outputs.

Secondly, previous evaluations of MicroTraitLLM noted the absence of a robust validation mechanism for assessing the quality of retrieved articles. The earlier approach relied on a heuristic LLM-based grading prompt, which asked the model to rate how effectively each article addressed the user's query. To establish a more consistent and reliable validation process, we integrated FlashRank, a lightweight reranking model capable of evaluating article relevance and journal quality [2]. FlashRank now scores all candidate articles retrieved during the MTLLM pipeline, enabling the system to prioritize higher-quality sources when generating responses. This enhancement provides a more rigorous, reproducible, and transparent basis for content selection within MicroTraitLLM.

Results:

Validation Studies

Model Comparison Study

The model comparison study evaluated four biomedical language models—BioGPT-finetuned, BioGPT, BioMedLM, and a GPT-2 baseline—across accuracy, performance, and computational efficiency within the MicroTraitLLM framework. All models were assessed using a 15-query benchmark spanning diverse microbiological domains and organism types. Each query required identification of microbial species and extraction of a comprehensive set of biological traits.

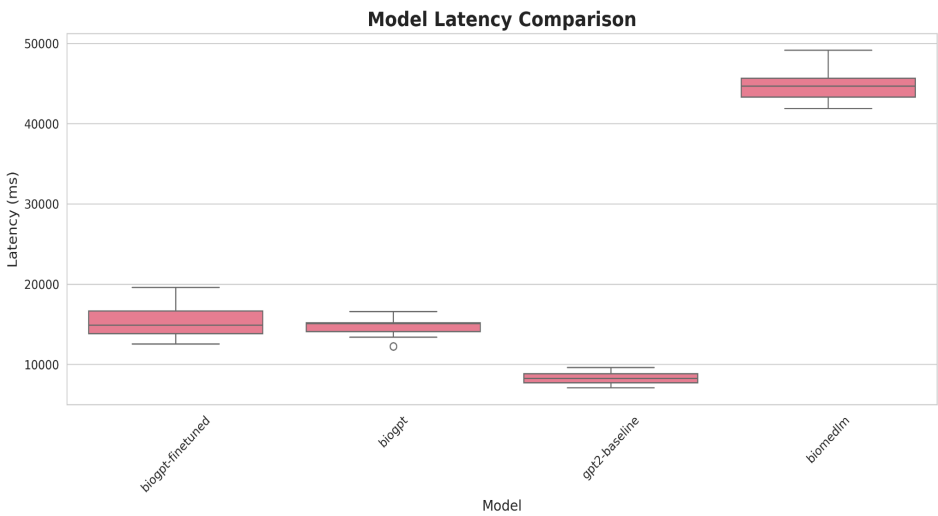


Figure 1: Latency Comparison by Model

The fine-tuned BioGPT model demonstrated the highest extraction accuracy, correctly identifying

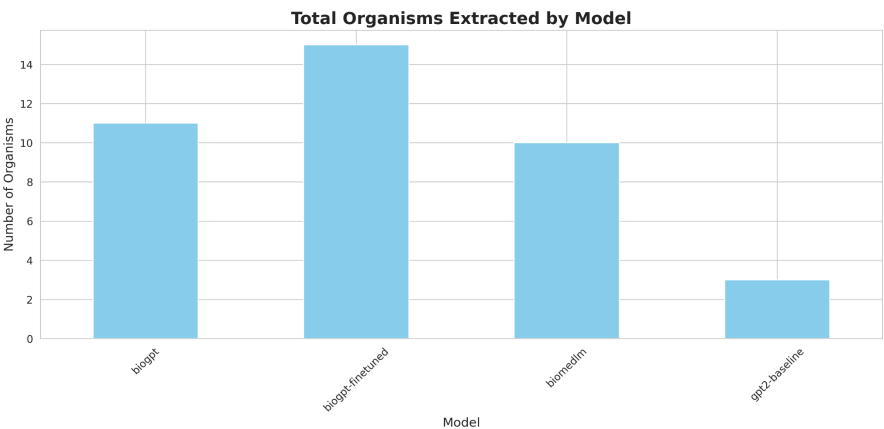


Figure 1: Extraction Accuracy by Model

15 organisms and recovering 100 biological traits, yielding a 100% success rate across all 15 queries (Table 1). Untuned BioGPT achieved moderate performance, identifying 11 organisms and 46 traits with a success rate of 73%. BioMedLM produced similar but slightly lower performance, extracting 10 organisms and 52 traits, with 67% successful extractions (See Figure 3). In contrast, the GPT-2 baseline

exhibited very limited extraction capability, recovering only 3 organisms and 7 traits (20% success rate). These results indicate that domain specialization and fine-tuning are critical for accurate microbial information extraction, with the fine-tuned BioGPT model outperforming all other models by a substantial margin (See Figure 1).

Latency measurements revealed substantial differences in speed across model sizes and architectures (See Figure 2). The GPT-2 baseline was the fastest model, with an average latency of 8,278 ms and low variability (± 830 ms). BioGPT and BioGPT-finetuned showed similar latency profiles, averaging 14,801 ms and 15,326 ms, respectively, with moderate variability. BioMedLM exhibited the slowest performance at 44,973 ms on average—approximately three times slower than the BioGPT variants.

Token usage mirrored model size and response verbosity. GPT-2 generated the fewest tokens overall (4,560; 304 per query), whereas BioMedLM produced the most (7,720; 515 per query). The BioGPT variants fell between these extremes. Higher-performing models tended to generate more tokens per query, consistent with more complete trait extraction and descriptive outputs.

To evaluate computational efficiency relative to extraction quality, we calculated organisms extracted per second and



Figure 3: Traits Extracted by Model

examined trade-offs between accuracy and speed. Although the GPT-2 baseline was the fastest model, its extremely low extraction accuracy resulted in the lowest efficiency when normalized by correct outputs. BioGPT-finetuned exhibited the highest accuracy-normalized efficiency, achieving perfect extraction performance at only modest latency costs. BioMedLM, despite recovering more traits than untuned BioGPT, showed the lowest overall computational efficiency due to substantially higher latency.

Model	BioGPT-finetuned	BioGPT	BioMedLM	GPT-2 baseline
Queries Processed	15	15	15	15

Organisms Extracted	15	11	10	3
Traits Extracted	100	46	52	7
Successful Extractions	15 / 15 (100%)	11 / 15 (73%)	10 / 15 (67%)	3 / 15 (20%)
Average Latency (ms)	15326.33	14800.84	44973.14	8278.21
Latency SD (ms)	2059.87	1124.18	2173.51	830.23
Median Latency (ms)	14881.05	15067.06	44689.20	8242.98
Total Tokens Used	7121	6642	7720	4560
Average Tokens/Query	474.7	442.8	514.7	304.0

Table 1. Summary of extraction accuracy, performance characteristics, and computational efficiency for all evaluated language models. The table reports the number of organisms and traits extracted, the proportion of successful queries, latency metrics (average, standard deviation, and median), and token usage for each model when applied to the 15-query microbiology benchmark dataset.

A clear speed–accuracy trade-off emerged across models. The GPT-2 baseline demonstrated that small, general-purpose models offer fast response times but fail to achieve usable extraction accuracy. Conversely, BioMedLM showed that increasing model size alone does not guarantee improved performance, as slower inference times reduced practical utility. The BioGPT-finetuned model offered the strongest balance: although slower than GPT-2, it achieved near-identical latency to untuned BioGPT while dramatically improving extraction accuracy across all metrics. This suggests that domain-specific fine-tuning provides greater performance gains than scaling model size within the MicroTraitLLM framework.

Overall, the results show that Biomedical fine-tuning is essential for accurate microbial trait extraction, with the fine-tuned BioGPT model outperforming all others in accuracy, efficiency, and robustness. While smaller models provide faster inference, they lack the capability needed to reliably extract microbial traits. Larger general biomedical models such as BioMedLM incur significant performance overhead without proportional accuracy benefits. The BioGPT-finetuned model therefore represents the most effective and computationally balanced model for integration into the MicroTraitLLM system.

Article Validation and Citation Accuracy (AVCA)

To evaluate the performance of the AVCA subsystem, we processed a random subset of 3,000 documents from the larger corpus of approximately 18,000 biomedical text files. Each document was parsed into its constituent elements, including title, abstract, and main text, and then vectorized using the shared TF-IDF model used for the user query. The resulting document–term matrix had a shape of 3000×2000 , reflecting the dimensionality of the vocabulary learned during preprocessing. This vector representation served as the basis for computing three similarity measures between each document and the test query “*antibiotic resistance in gram-negative bacteria.*”

The statistical summary of similarity scores is shown in Table 2. Cosine similarity and dot product similarity displayed identical minimum, maximum, and mean values, which is expected given that the dot product reduces to a scaled version of cosine similarity when vectors are normalized by TF-IDF magnitudes. Pearson correlation exhibited a slightly lower mean value and a similar overall range, reflecting its sensitivity to proportional rather than absolute term usage patterns. The standard deviation across all metrics was comparable, indicating a consistent degree of variability in document–query alignment across methods.

Table 2. Summary statistics for similarity metrics.

Metric	Min	Max	Mean	Std. Dev.
Cosine similarity	0.0031	0.3093	0.0530	0.0537
Dot product similarity	0.0031	0.3093	0.0530	0.0537
Pearson correlation	−0.0060	0.3053	0.0434	0.0547

Figure 4. Distribution of cosine similarity scores between the query and 3,000 sampled articles.

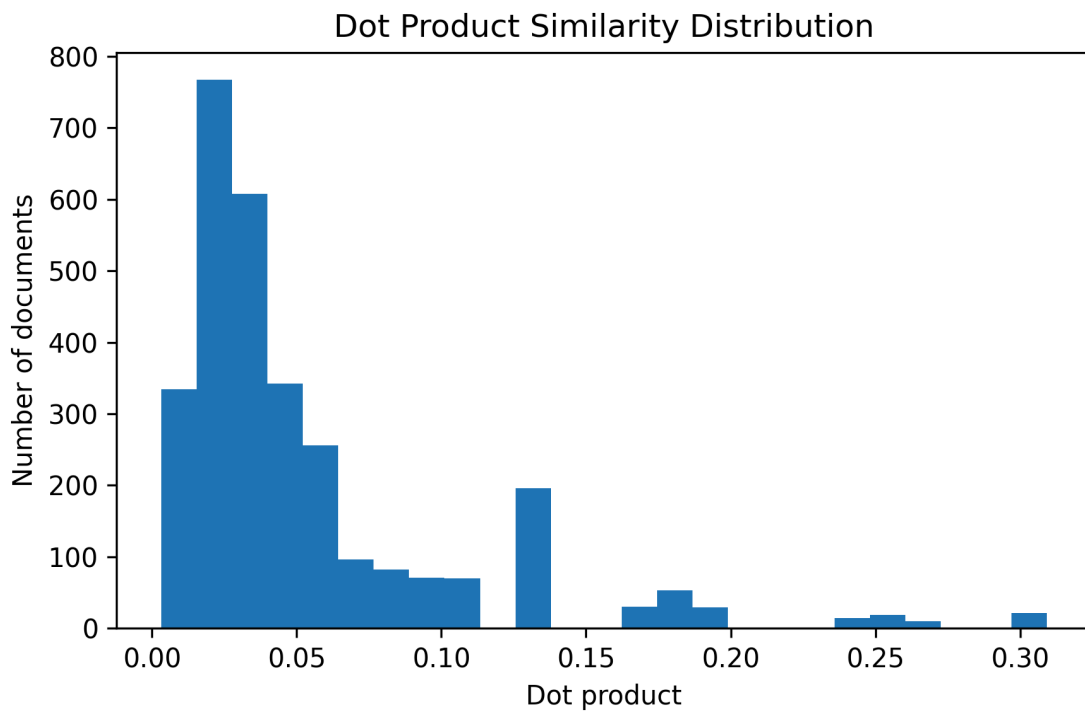
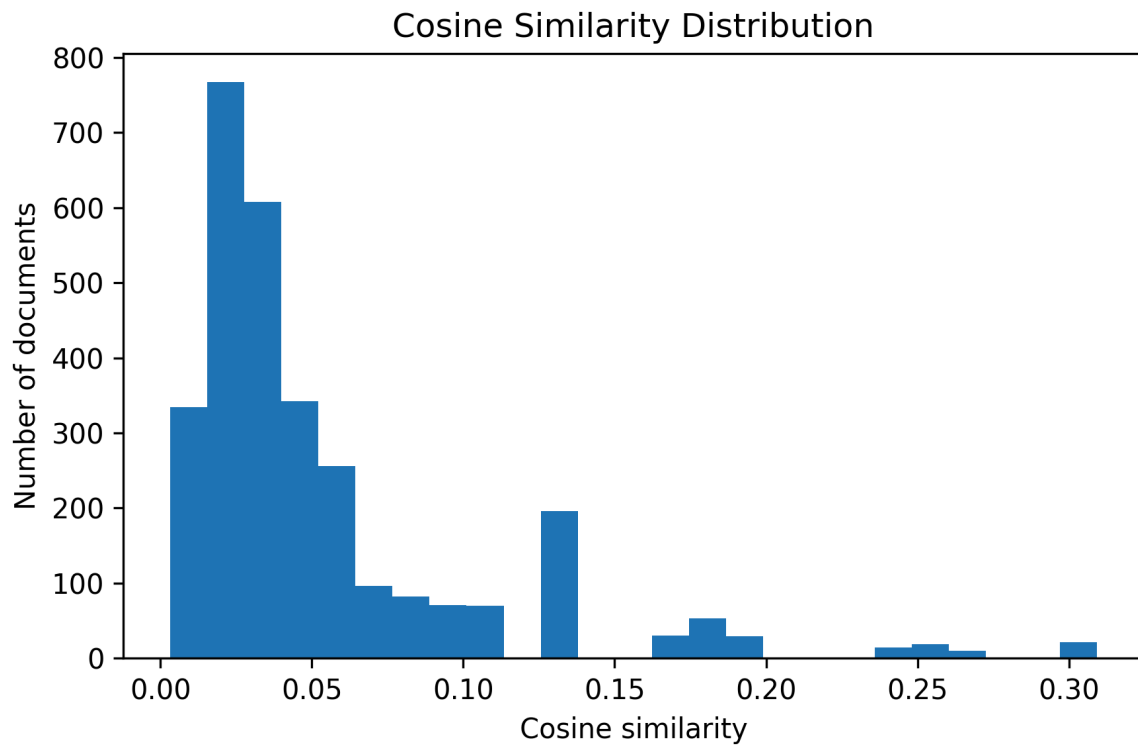


Figure 5. Distribution of dot product similarity scores demonstrating similar relevance structure.

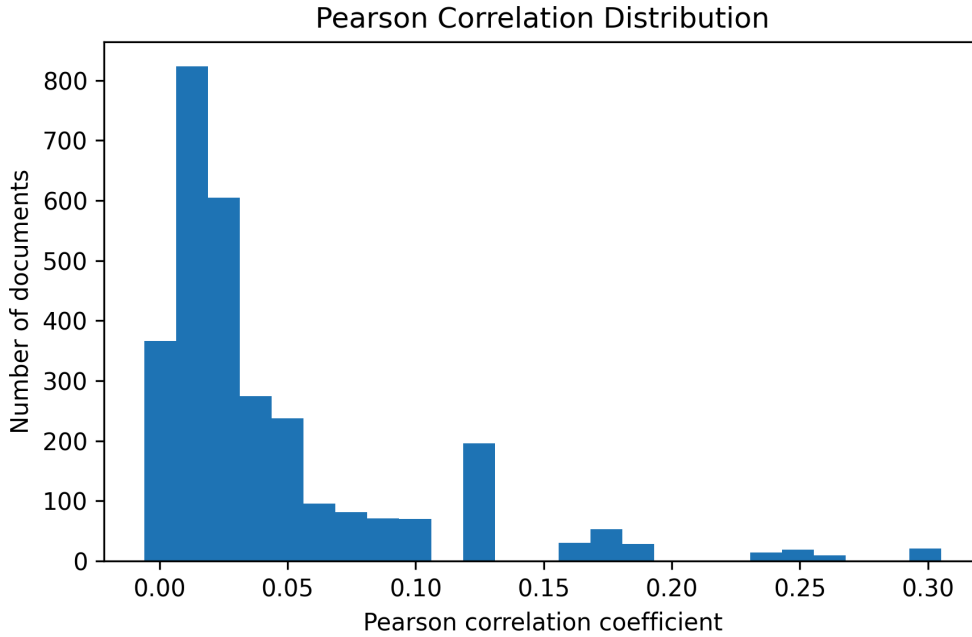


Figure 6. Distribution of Pearson correlation coefficients showing proportional term alignment.

The distribution of similarity values for each metric was visualized using histograms (Figures 4–6). All three metrics produced right-skewed distributions, with the majority of documents exhibiting low to moderate similarity to the query and a smaller number demonstrating high alignment. The near-identical shape of these histograms suggests that the TF-IDF embedding space encodes a strong and consistent semantic signal, such that cosine similarity, dot product similarity, and Pearson correlation effectively identify overlapping groups of relevant documents. This convergence across metrics reinforces the stability of AVCA’s relevance ranking.

To understand how AVCA segments documents into actionable relevance categories, we applied quantile-based banding to each similarity metric, dividing results into Low, Medium, and High confidence groups. Table 2 presents the distribution of documents across these bands. The near-uniform proportions across cosine, dot product, and Pearson correlation demonstrate that confidence assignments are not sensitive to the choice of similarity measure. This robustness ensures that AVCA can be reliably integrated into downstream system components without concern for metric-specific behavior.

Table 2. Proportion of documents in each relevance band.

Metric	Low %	Medium %	High %
Cosine	32.8%	32.7%	34.5%
Dot Product	32.8%	32.7%	34.5%
Pearson	32.8%	33.0%	34.2%

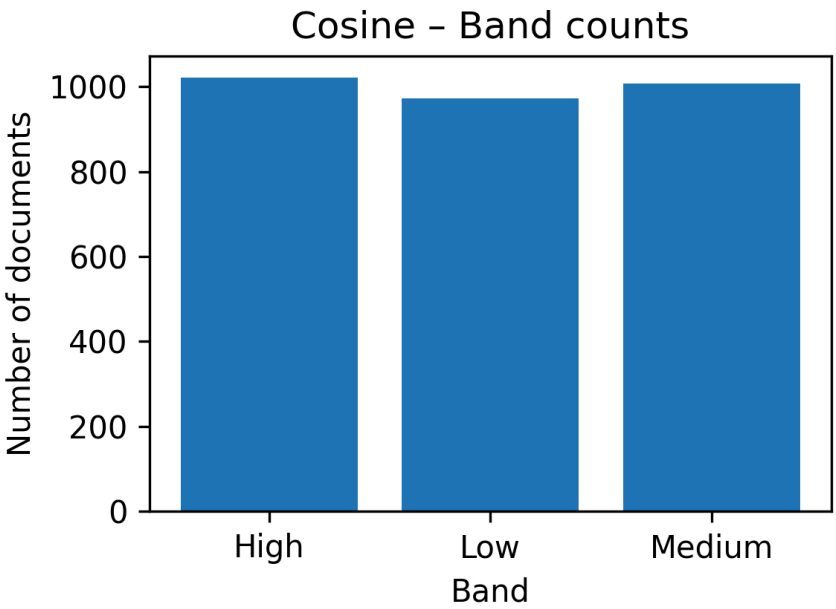


Figure 7. Confidence band distribution for cosine similarity metric using percentile thresholds.

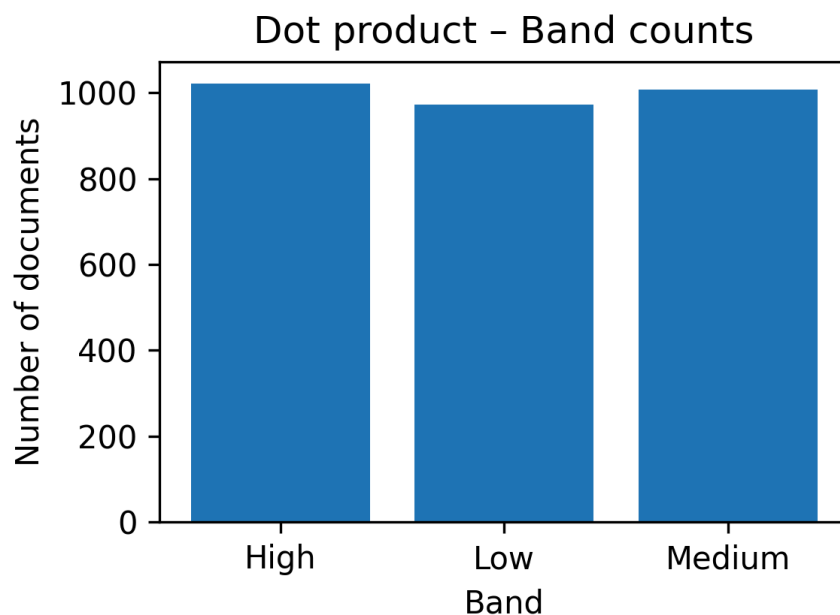


Figure 8. Confidence band distribution for dot product similarity metric using percentile thresholds.

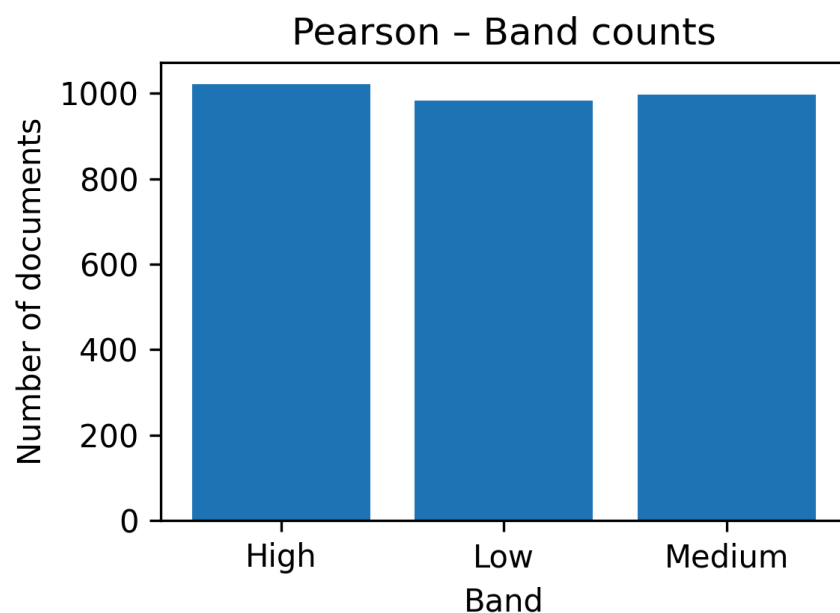


Figure 9. Confidence band distribution for Pearson correlation metric using percentile thresholds.

Visualizations of the band distributions (Figures 7–9) further confirm that all three similarity metrics classify documents into nearly identical relevance tiers. This outcome is consistent with the statistical summary and histogram analysis: because all metrics operate on the same TF-IDF vector space,

and because the TF-IDF representation itself strongly encodes topic-specific vocabulary, the ranking behavior is extremely stable. The slight divergence observed for Pearson correlation can be attributed to its emphasis on the proportional relationship between feature distributions, but the difference is minimal in practice.

Overall, these results indicate that the AVCA subsystem reliably identifies highly relevant documents and does so consistently across multiple similarity metrics. The combination of TF-IDF embedding, similarity scoring, and quantile-based banding provides a stable and interpretable foundation upon which later components—such as citation validation and LLM response generation—can depend. This positions AVCA as a robust module ready for integration into the larger system pipeline.

Vector Ablation Study

We conducted an ablation study to determine the best number of top K vectors to use when answering complex biological questions in the microbial domain (See Figures 10 and 11). Both Llama3.2 and Gemma2 models were evaluated separately. The Top K vectors for Llama3.2 were at K=1 and K=2,

which gave hybrid accuracy scores of 0.12. The Top K vectors for Gemma2 were at K=2 and K=5, which both gave a hybrid accuracy score of 0.13. The standard deviation for all Top K measurements approximated at about 0.06. Interestingly, under this methodology, there appeared to not be much difference between the vectors. This could potentially be attributed to the questions not being

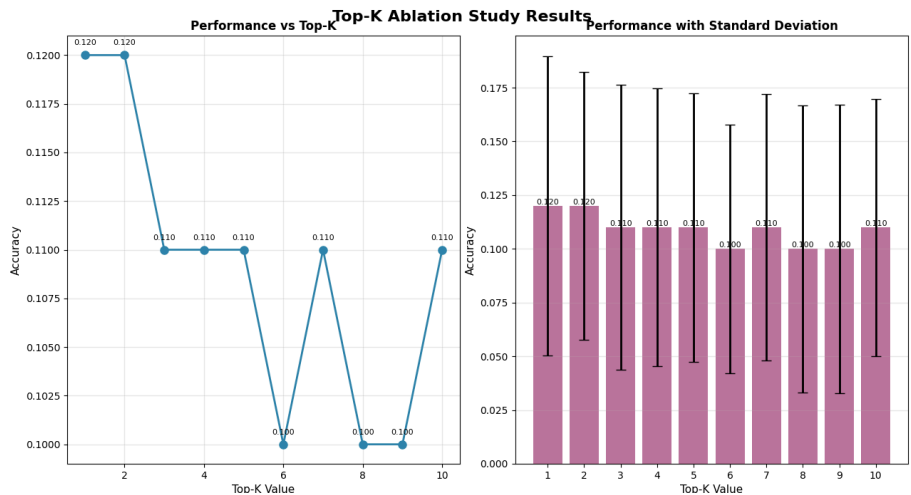


Figure 10: Top-K Results for Llama3.2

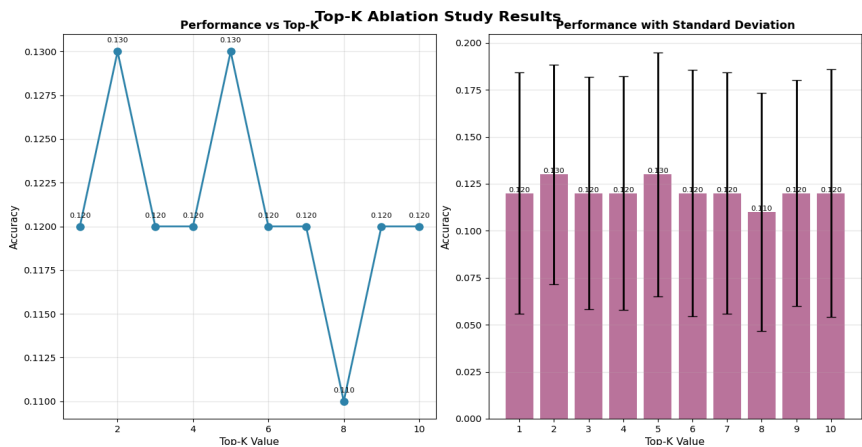


Figure 11: Top-K Results for Gemma2

long enough to constitute proper corpora for comparison.

We also attempted another analysis format; however, this time we attempted to look for only key words or phrases within an answer that could potentially constitute a correct answer (See figures 12 and 13). Here, the answers were scored out of how many key words were identified, ($\frac{1}{3}$ correct, .33 evaluation score). Although not a robust evaluation score, Figures 12 and 13 highlight a more important story, where MicroTraitLLM obtains vector scores in a much more expected manner. Here, the top-K for Llama3.2 was K=8 at an accuracy score of .34, whereas the top-K for Gemma2 was K=10 at an accuracy score of 0.38. However, this methodology also yielded a much higher standard deviation, with most values around 0.26. Although this methodology likely could not be used for publication efforts, it does demonstrate that MicroTraitLLM is obtaining key information for the user query.

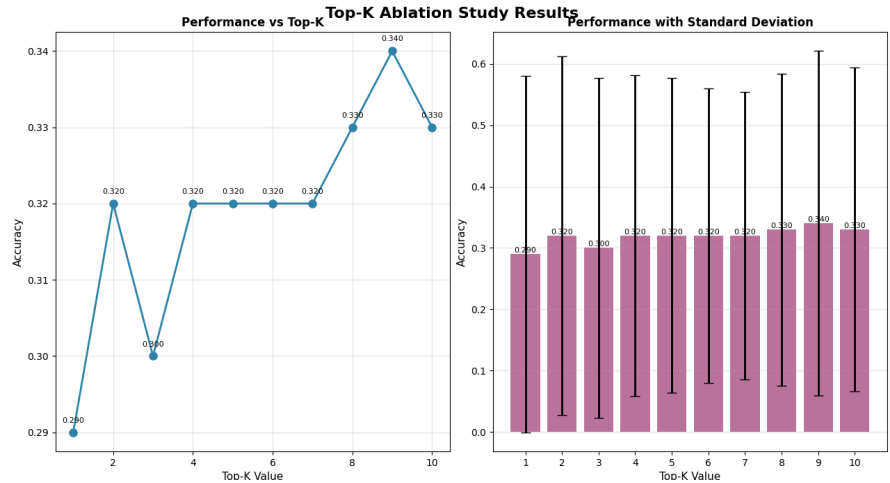


Figure 12: Top-K for Llama3.2 using Keywords

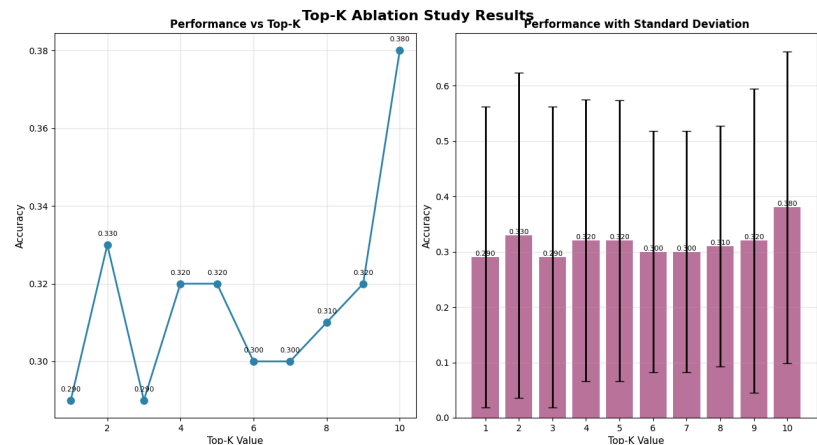


Figure 13: Top-K for Gemma2:2b using Keywords

New Integrations

Low Rank Adaption (LoRA) Fine Tuning

The results of the Mistral Fine tuning produced a model that gave an answer within text citations in a concise, academic tone, without any other fluff produced by an untuned LLM. Mistral (prior to fine tuning) was comparable to other models with a similar parameter range, such as GPT4o-mini. Both models had a tendency to create extra “fluff” in their responses when prompted using system prompting. This resulted in things such as an extraneous reference footer rather than in-line citations.

However, after fine tuning, a distinct change in output can be noticed compared to the default responses provided by an untuned model such as GPT 4o-mini. The structure is more academic, hallucinations are rarer, and extraneous output is nonexistent.

To evaluate the fine tuned model, an evaluation script was produced to check the context for key words, as well as check for in text citations, as well as no extraneous responses (e.g. the model responding with “Sure thing!” prior to giving the response) For example, if the question asks about common bacteria that causes upper respiratory infections, *Staphylococcus aureus* should be mentioned somewhere in the answer.

Database Optimization

We evaluated the improvement of MicroTraitLLM response time using the local database versus previous iterations, which depended on external database API calls. As can be seen in Figure 14, the local database drastically cuts down on response time across models. Average response time for the local database (LocalDB) method sat at about 5.66 seconds, whereas

Llama3.2 with an external database took approximately 19.52 seconds, and Gemma2 with an external database took approximately 15.24 seconds. This is ideal for future work, as we can continue to provide competitive response times to larger corporation models as well as evaluate alternative methodologies, such as new models, in a faster time frame than we could accomplish previously.

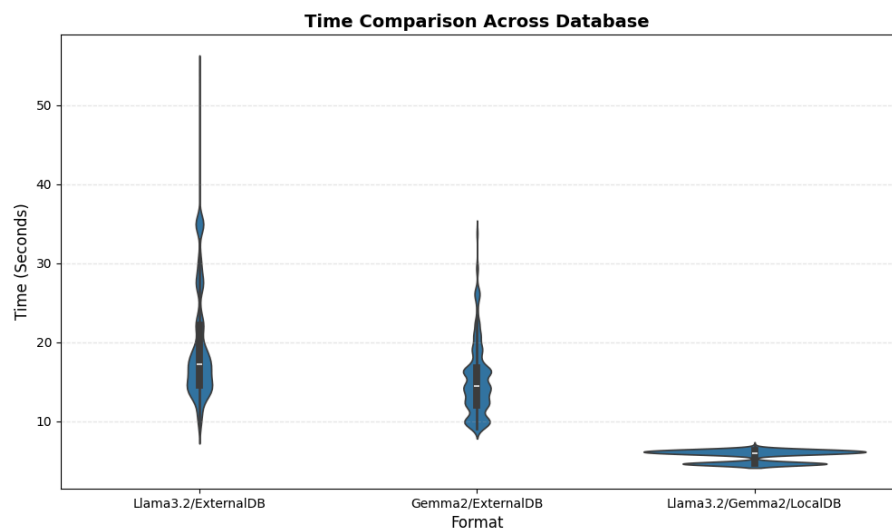


Figure 14: Violin plot comparing Time to Response (s)

Discussion:

Overall, we demonstrate many key results in this work. Firstly, through our AVCA validation studies, we demonstrate that MicroTraitLLM is able to discuss relevant topics based on the user query. Our top K ablation study results also reinforce this idea, while also showing that the system continues to produce answers that are in line with expected answers in the microbial domain. Secondly, our model comparison study reveals that BioGTP-finetuned could serve as an alternative, even smaller model than what is already used in MicroTraitLLM. Finally, our brief study into response time per database archetype shows that the local database storage system we integrated has dramatically sped up our time to response when compared to the external database-oriented system we were using earlier. Overall, these improvements lead to a greater trust in MicroTraitLLM due to the studies and implementations we have created.

However, our work still has many avenues for exploration. Firstly, we need to continue to verify the efficacy of our VQA system. We were unable to test on microbe-specific data, so we cannot say right now whether we can accurately answer user questions pertaining to visual data. We were also unable to verify the efficacy of our newly fine-tuned Mistral model using LoRA. Though we saw positive results, with the model providing structured responses, we were unable to quantify improvements over alternative models or the original version of MicroTraitLLM. Both of these systems would benefit greatly from this evidence. Secondly, MTLLM would benefit from a deeper dive into hallucination quantification. Rather than attempting to demonstrate that the answers provided by MTLLM are “correct”, showing evidence that MTLLM hallucinates less and provides direct evidence from the articles it uses may be a more effective methodology.

Another important direction concerns hallucination detection and mitigation. Rather than focusing solely on whether responses are “correct,” future work should measure the frequency, severity, and nature of hallucinations in MicroTraitLLM’s outputs. Approaches such as citation-attribution scoring, fact-consistency benchmarks, and automated hallucination classifiers could help evaluate how reliably the model anchors its answers in the underlying literature. Demonstrating reduced hallucination rates, as well as continuing to provide transparent evidence for model answers, would substantially strengthen user confidence in MicroTraitLLM and its capabilities.

More broadly, our findings contribute to growing evidence that domain-adapted LLM systems can serve as valuable tools for microbe researchers. The improved retrieval structure, faster response times, and emerging fine-tuning strategies collectively indicate that the MicroTraitLLM framework, or another similar RAG-based system, may soon integrate more deeply into scientific workflows, such as accelerating literature reviews or assisting in experimental design. However, to reach this goal, MicroTraitLLM must overcome its current limitations: the need for further curated evaluation datasets, the challenge of quantifying “correct” answers, and the broader question of how to best align LLM outputs with scientific standards of evidence.

In summary, while MicroTraitLLM advances the state of LLM-assisted microbial trait analysis, the results also highlight several future research directions. Achieving robust VQA performance, refining fine-tuned model architectures, systematically quantifying hallucination behavior, and developing

rigorous benchmarking frameworks are all essential methods for building the next generation of trustworthy, domain-grounded language models for the microbe domain. Continued work in these areas will not only strengthen MicroTraitLLM but also contribute to the broader effort to integrate LLMs responsibly and effectively into microbe-focused research.

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